

React & Node.js Assignment Sheet

Assignment 1: React - Student Course Manager

Objective:

Build a React application to manage courses without any backend.

Requirements:

1. Create components: App, Navbar, CourseForm, CourseList, CourseCard.
2. Add a new course (Title, Trainer, Duration).
3. Validate empty fields.
4. Display all courses using list rendering with unique keys.
5. Search courses by title.
6. Filter courses: All / Beginner / Intermediate / Advanced.
7. Mark course as Completed.
8. Delete a course.
9. Show total courses and completed courses.
10. Persist data using localStorage.
11. Use useState for state management.
12. Use useEffect to load and save localStorage.

Bonus:

- Dark Mode
- Responsive UI
- Edit existing course

Assignment 2: Node.js + Express - Library Management API

Objective:

Create REST APIs using Express. Store data in books.txt (or library.txt). MongoDB is NOT allowed.

Book Object:

id, title, author, category, price, available

Required APIs:

GET /books

GET /books/:id

POST /books

PUT /books/:id

PATCH /books/:id

DELETE /books/:id

GET /books/search?title=

GET /books/category/:category

Rules:

- Read and write using fs module.
- Data must persist inside text file.
- Validate required fields.
- Return proper HTTP status codes.
- Handle invalid IDs.
- Organize project into routes, controllers, utils/file helper.

Folder Structure:

```
project/  
  routes/  
  controllers/  
  utils/  
  books.txt  
  server.js
```

Bonus:

- Sorting by price
- Pagination
- Logging middleware
- Request timestamp middleware